

Bruhat order and applications: Lecture 2

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Today we will focus on the weak order of a Coxeter group W . We will see that it is heavily tied both to inversions and to geometry.

1 Weak order

Let's recall the definition.

Definition 1.1. Let W be a Coxeter group with simple reflections S . The **weak order** is the partial order on W denoted by $\leq_R$ which puts

$$u \leq_R v$$

if and only if there exists a reduced word $s_{i_1}s_{i_2}\cdots s_{i_k}$ for v and an index $0 \leq j \leq k$ so that

$$u = s_{i_1}\cdots s_{i_j}.$$

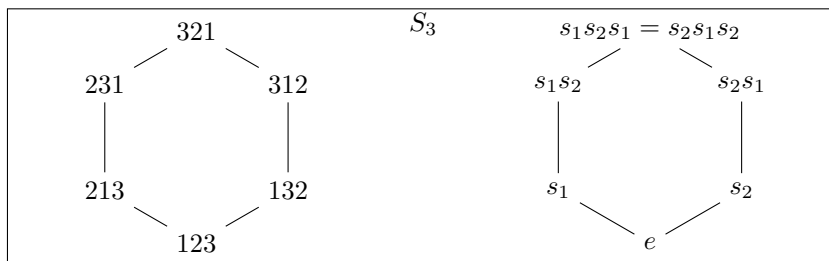
In other words, $u \leq_R v$ if there is some reduced word for u which is a left-justified subword (also called a **prefix**) of some reduced word for v . As an example in S_3 , we can see that $213 \leq_R 231$ since the reduced word s_1 is a left-justified subword of s_1s_2 . On the other hand, $132 \not\leq_R 231$ since there is only one reduced word for 231 , which is s_1s_2 , and it does not have s_2 as a prefix.

If we have a reduced word $s_{i_1}s_{i_2}\cdots s_{i_k}$ for v , then we can list off $k+1$ many elements that are $\leq_R v$:

$$e, s_{i_1}, s_{i_1}s_{i_2}, s_{i_1}s_{i_2}s_{i_3}, \dots, s_{i_1}s_{i_2}\cdots s_{i_k}.$$

If $s_{i_1}s_{i_2}\cdots s_{i_k}$ is the only reduced word for v , then these $k+1$ elements are all of the elements $\leq_R v$. But in general, you would need to do this for every reduced word of v . For example in S_3 , there is the inequality $s_2 \leq_R s_1s_2s_1$ because the reduced word $s_1s_2s_1$ is equal to the reduced word $s_2s_1s_2$, which has s_2 as a prefix.

Note that the identity e is the “empty subword” of any word, so $e \leq_R v$ for all $v \in W$. With all this in mind, we can write down the Hasse diagram for the weak order on S_3 . Luckily, we will soon learn much easier ways of understanding weak order.



We will see in the exercises that if v covers u in weak order, then $v = us$ for some $s \in S$, and $\ell(v) = \ell(u)+1$. In other words, cover relations change the length by exactly 1. This means that when we draw the Hasse diagram, we can put elements with the same length on the same horizontal line. Figure 1 shows the weak order on S_4 , drawn in this way.

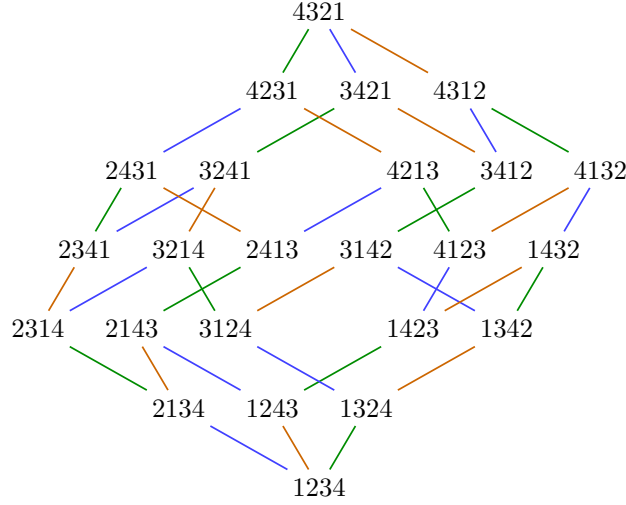
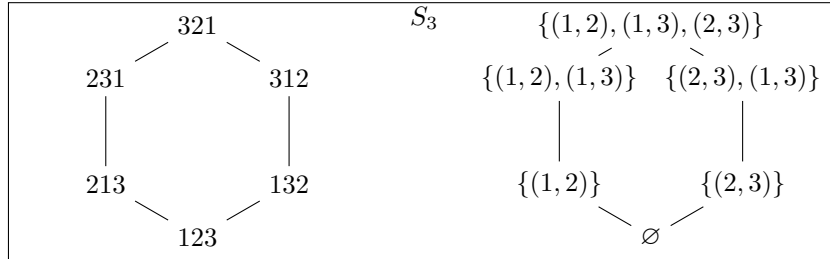


Figure 1: Weak order on S_4 . Cover relations coming from right-multiplying by s_1 are colored blue, by s_2 are colored green, and by s_3 are colored orange.

2 Inversion sets

The definition of weak order in terms of reduced words can be tricky to work with. It will be extremely useful to reframe this definition in terms of inversions. This will give us a way to check the weak order relation using inversion sets. To see the idea, look at the inversion sets of the permutations in S_3 .



If w is an element of W and $X \subseteq T$ is a set of reflections, then we use the notation wXw^{-1} to mean $\{wtw^{-1} \mid t \in X\}$. Note that wXw^{-1} is also a set of reflections.

Lemma 2.1. *Let w be an element of a Coxeter group W , and let $s \in S$. Then the following hold:*

- (a) *If $\ell(ws) < \ell(w)$, then $\text{inv}(w) = \text{inv}(ws) \cup \{ws w^{-1}\}$. Moreover, $ws w^{-1}$ is not in $\text{inv}(ws)$.*
- (b) *If $\ell(sw) < \ell(w)$, then $\text{inv}(w) = s \text{inv}(sw)s \cup \{s\}$. Moreover, s is not in $s \text{inv}(sw)s$.*

Proof. Assume $\ell(ws) < \ell(w)$, and let $w' = ws$. Let $s_{i_1} \cdots s_{i_k}$ be a reduced word for w' . Then $s_{i_1} \cdots s_{i_k} s$ is a reduced word for w . The standard inversion sequence for this word tells us that the inversions of w are

$$s_{i_1}, s_{i_1} s_{i_2} s_{i_1}, \dots, s_{i_1} \cdots s_{i_k} s s_{i_k} \cdots s_{i_1}.$$

The last inversion in this list is $ws w^{-1}$. The remaining inversions are exactly the elements of $\text{inv}(w')$. This proves (a).

Assume $\ell(sw) < \ell(w)$, and let $w' = sw$. Let $s_{i_1} \cdots s_{i_k}$ be a reduced word for w' . Then $s s_{i_1} \cdots s_{i_k}$ is a reduced word for w . The standard inversion sequence for this word tells us that the inversions of w are

$$s, s s_{i_1} s, s s_{i_1} s_{i_2} s_{i_1} s, s s_{i_1} \cdots s_{i_k} \cdots s_{i_1} s.$$

The first inversion is s , and the rest are exactly the elements of $\{sts \mid t \in \text{inv}(w')\}$. This proves (b). $\square$

Let us observe the power of Lemma 2.1(a). We already knew that all cover relations in weak order are of the form $ws \lessdot w$ for some $s \in S$. (We use the notation $u \lessdot v$ to mean that v covers u .) Now the lemma tells us that we can describe what happens to the inversion set in this situation: the inversion set $\text{inv}(w)$ is built by adding a single inversion to $\text{inv}(ws)$, and that inversion is $ws w^{-1}$. Hence, if we are looking at the Hasse diagram of weak order, then we can think of going down along an edge of the diagram from w as either

- Right-multiplying w by a descent s , or
- Removing the reflection $ws w^{-1}$ from $\text{inv}(w)$.

Since $u \leq_R v$ if we can get to u from v by going down along Hasse diagram edges, it follows that the inversion set of u is obtained by deleting reflections from the inversion set of v . Based on this reasoning, we could hope that we can check whether $u \leq_R v$ by simply checking that the inversion set of u is contained in the inversion set of v . This turns out to be true.

Theorem 2.2. *Let u and v be elements of a Coxeter group W . Then $u \leq_R v$ if and only if $\text{inv}(u) \subseteq \text{inv}(v)$.*

Proof. Assume $u \leq_R v$. Then there is a reduced word $s_{i_1} \cdots s_{i_k}$ for v and an index $0 \leq j \leq k$ such that $u = s_{i_1} \cdots s_{i_j}$. This word is in fact a reduced word for u . Hence by the definition of a Coxeter group, every inversion of u is of the form $s_{i_1} \cdots s_{i_{r-1}} s_{i_r} s_{i_{r-1}} \cdots s_{i_1}$ for some $r \leq j$. These elements are also evidently inversions of v . It follows that $\text{inv}(u) \subseteq \text{inv}(v)$.

We will prove that $\text{inv}(u) \subseteq \text{inv}(v)$ implies $u \leq_R v$ by induction on $\ell(u)$. If $\ell(u) = 0$, then $u = e$ and $e \leq_R v$. Otherwise, assume $\ell(u) > 0$ and $\text{inv}(u) \subseteq \text{inv}(v)$, and that we know $\text{inv}(u') \subseteq \text{inv}(v')$ implies $u' \leq_R v'$ whenever $\ell(u') < \ell(u)$. Then u^{-1} has a descent s , since it is not the identity. Equivalently, $s \in \text{inv}(u)$. By assumption, s is also in $\text{inv}(v)$. Hence by Lemma 2.1(b),

$$\text{inv}(u) = s \text{inv}(u')s \cup \{s\}$$

$$\text{inv}(v) = s \text{inv}(v')s \cup \{s\},$$

where we have introduced the notations $u' = su$ and $v' = sv$. These equalities together with the fact that $\text{inv}(u) \subseteq \text{inv}(v)$ (and the fact that s is not in $s \text{inv}(u')s$) imply that $s \text{inv}(u')s \subseteq s \text{inv}(v')s$. And this further implies that $\text{inv}(u') \subseteq \text{inv}(v')$. Since $\ell(u') < \ell(u)$, we can use the inductive hypothesis to deduce that $u' \leq_R v'$. Hence there is a reduced word for v' , say $s_{i_1} \cdots s_{i_k}$, which has a prefix that is a reduced word for u' , say $s_{i_1} \cdots s_{i_j}$. We now use the fact that $u = su'$ and $v = sv'$ to deduce that $ss_{i_1} \cdots s_{i_j}$ is a reduced word for u which is a prefix of the reduced word $ss_{i_1} \cdots s_{i_k}$ for v . Hence, $u \leq_R v$. By induction, this proves the theorem. $\square$

Corollary 2.3. *Two elements $u, v \in W$ are equal if and only if $\text{inv}(u) = \text{inv}(v)$.*

Proof. By the theorem, if $\text{inv}(u) = \text{inv}(v)$, then we have both $u \leq_R v$ and $v \leq_R u$. Since weak order is a partial order, this implies that $u = v$. $\square$

3 Coxeter complex

Now we are going to turn weak order into some geometry. There are lots of ways to do this. Our first entry point is called the Coxeter complex. To define it, let's focus on rank 3 Coxeter groups. In this case, the Coxeter complex is built from triangles. Each element of W has an associated triangle. We label the edges of this triangle by the three simple reflections. Then we glue together triangles. The triangle of w is glued to the triangle of w' along the edge labeled by s if $ws = w'$. At the end, we have an object built from triangles. For S_4 (which, remember, has rank 3), the Coxeter complex looks something like Figure 2. In that figure, we have not yet glued all the triangles. That is because doing so would require bending it in 3d. If we finish gluing, then we get the result shown in Figure 4. As the figure shows, what we get is actually a triangulation of the sphere. If you keep the triangles flat, then you can arrange so you get a triangulation of the surface of a tetrahedron (shown in Figure 3).

Now that we've placed permutations in 3-dimensional space, we can see what shape the Hasse diagram makes. If we draw the edges from the Hasse diagram, we get the object in Figure 5. We can imagine filling in the polygons in that shape. The resulting object is a polyhedron called the **permutahedron**, shown in Figure 6.

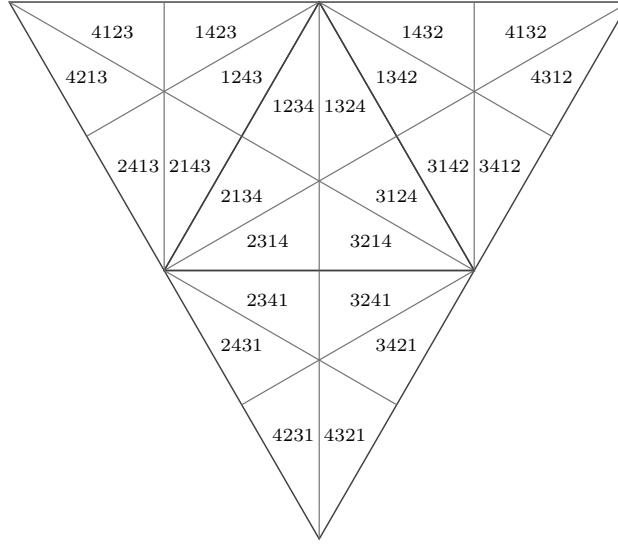


Figure 2: The Coxeter complex for S_4 , with some of the edges left unglued so it can lie flat. Each triangle is labeled by the corresponding element of S_4 .

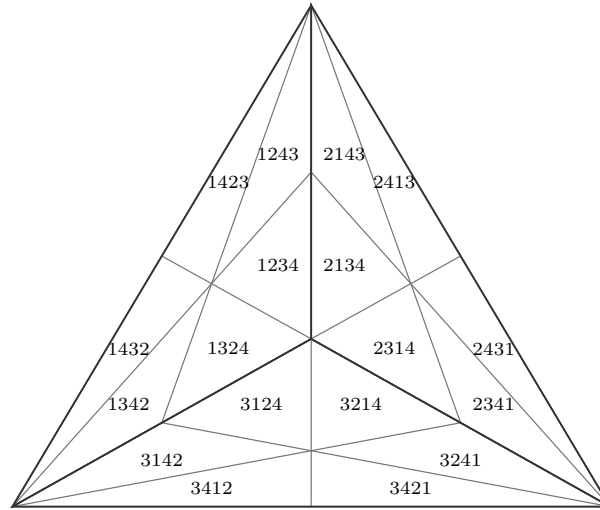


Figure 3: The Coxeter complex of S_4 can be viewed as a triangulation of the surface of a tetrahedron.

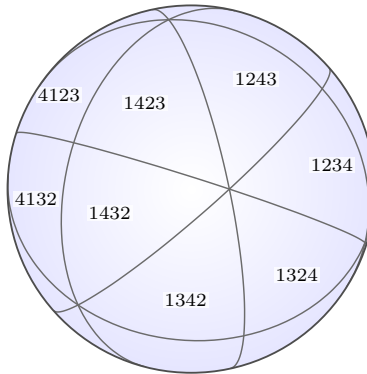


Figure 4: The Coxeter complex for S_4 can be viewed as a triangulation of the sphere. (This is what we get by inflating the tetrahedron like a balloon.)

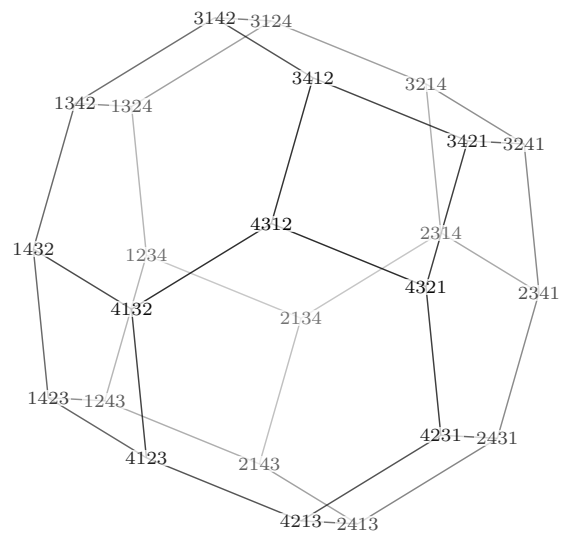


Figure 5: The Hasse diagram for weak order on S_4 , embedded in 3-dimensional space.

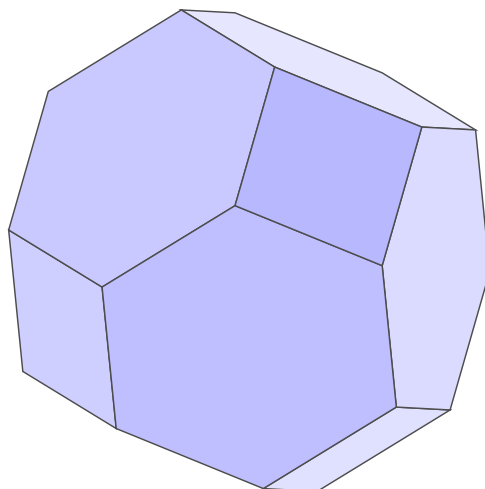


Figure 6: The permutahedron for S_4 .

When you examine Figure 4, one pattern that emerges is that the edges of the triangles actually connect together to form great circles on the sphere. (In a random triangulation of the sphere, this won't be the case.) This hints at another way of building the Coxeter complex, one that is more geometric. This way passes through an “arrangement of planes”. (The common term for these is **hyperplane arrangement**; in 3D, a hyperplane is the same thing as a plane. In 2D, a hyperplane is a line.) Recall that a plane in 3D is the set of solutions to a linear equation. To build the Coxeter complex for S_4 , we use 6 planes. The linear equations we use are actually easier to state using 4 variables: x_1, x_2, x_3, x_4 . We eliminate x_4 using the equation $x_1 + x_2 + x_3 + x_4 = 0$. The 6 equations then are

$$x_1 = x_2, x_1 = x_3, x_1 = x_4, x_2 = x_3, x_2 = x_4, x_3 = x_4.$$

Note that these correspond with the reflections $(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4) \in T$. This set of 6 planes is called the **braid arrangement** because it is related to the “braid group”. We can get the Coxeter complex from it by intersecting the braid arrangement with the unit sphere. Each plane intersects the sphere in a great circle, and the 6 great circles you get are exactly the 6 great circles appearing in Figure 4. Check out gtbarkley.org/cmnd/ for an interactive visualization of this.

Let us give one last interpretation of these pictures. When we think about the edges of triangles in the Coxeter complex, we were previously thinking about the simple reflection we were using to right multiply. But if $w' = ws$ are in neighboring triangles, then we know from Lemma 2.1(a) that the inversion sets of w' and w differ by just one reflection. So there is a second way we can label the edge of a triangle: by labeling it with the reflection that is toggled from the inversion set. It turns out that if we do this, then every edge with the same reflection label is on the same great circle. In other words, this gives a correspondence between the planes of the arrangement and reflections. Crossing a plane corresponds to adding or removing that reflection as an inversion. Visit the link above for a demo showing this.

For a general Coxeter group W , we can still write down the Coxeter complex in the same way (using $(n - 1)$ -dimensional simplices instead of triangles, where n is the rank). When W is finite, it will still be a triangulation of the sphere. There will still be a hyperplane arrangement producing the Coxeter complex. This arrangement is called the **Coxeter arrangement**. It is built by using a **root system** for W . For S_4 , the root system is the set of vectors

$$\mathbf{e}_1 - \mathbf{e}_2, \mathbf{e}_1 - \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_4, \mathbf{e}_2 - \mathbf{e}_3, \mathbf{e}_2 - \mathbf{e}_4, \mathbf{e}_3 - \mathbf{e}_4,$$

in $\mathbb{R}^4$, as well as their negations. Note that these are normal vectors to the planes in the arrangement! The root system turns out to have many of its own applications. In a way, root systems are the gateway connecting Coxeter groups to applications in algebra, such as Lie groups and quiver representations. They could get a whole course on their own.

4 Problems

For some of these problems, you may want to use the visualization for S_3 and S_4 available at gtbarkley.org/cmnd/.

1. Verify that, in the symmetric group S_4 , the elements $s_2s_1s_3s_2$ and $s_2s_1s_2$ are incomparable in weak order.
2. Prove that weak order on a Coxeter group is a partial order.
3. Prove from the definition of weak order that if $u \prec_R v$ is a cover relation, then $\ell(u) = \ell(v) - 1$ and there exists some $s \in S$ so that $us = v$.
4. Draw the Hasse diagram for weak order on the dihedral group $I_2(4)$.
5. Let W be a Coxeter group of rank n . Show that for the weak order on W , the number of elements that w covers plus the number of elements that cover w is equal to n .
6. We say that a permutation π has a **312-pattern** if there exist $a < b < c$ so that $\pi^{-1}(c) < \pi^{-1}(a) < \pi^{-1}(b)$. (This just means that the three numbers appear in the order “biggest then smallest then middle”.) We say that π is **312-avoiding** if it has no 312 patterns.

- (a) How many 312-avoiding permutations are there in S_3 ? S_4 ? (Bonus problem: How many in S_n ?)
- (b) Let X_n be the set of 312-avoiding permutations in S_n . The weak order turns X_n into a poset called the **Tamari lattice**. Draw the Tamari lattice X_3 . (Time-consuming bonus: draw X_4 .)

The Tamari lattice X_n is related to cluster algebras. There is a bijection between X_n and the seeds of the type A_{n-1} cluster algebra. The graph you get by drawing an edge between two seeds related by a mutation turns out to be the same as the Hasse diagram for X_n . This graph also appears as the edge graph of a polytope called the **associahedron**. You can actually use the weak order on other Coxeter groups to understand other (finite-type) cluster algebras in a similar way, using **Cambrian lattices**.

7. Describe when two permutations of S_4 are on the same face of the simplex model for the Coxeter complex. Describe when two permutations of S_4 are on the same face of the permutahedron. What do you think the generalization is to S_n ?
8. Using the braid arrangement model for S_4 , verify that if a straight line segment is drawn between the permutation w and the identity permutation, then the planes that it passes through are exactly the inversions of w .
9. In this problem, assume that W is a Coxeter group with finitely many elements.
 - Show that there is an element $w_0 \in W$ such that, for all $w \in W$, we have $w \leq_R w_0$.
 - Show that, for all $w \in W$, the inversion set $\text{inv}(ww_0) = T \setminus \text{inv}(w)$.
 - Show that if $w \in W$ is an element such that $S \subseteq \text{inv}(w)$, then $w = w_0$.
 - Since every element of W is a prefix of some reduced word of w_0 , there must be quite a few reduced words for w_0 . Find three different reduced words for w_0 in S_4 .
10. The following problems are about inversions in the symmetric group S_n .
 - (a) Prove that if π is a permutation, then the inversion set of π has the following two properties
 - (i) If $1 \leq a < b < c \leq n$, and both transpositions (a, b) and (b, c) are in $\text{inv}(\pi)$, then $(a, c) \in \text{inv}(\pi)$.
 - (ii) If $1 \leq a < b < c \leq n$, and neither transposition (a, b) nor (b, c) is in $\text{inv}(\pi)$, then (a, c) is not in $\text{inv}(\pi)$.
 - (b) (Optional) Prove that if X is a set of transpositions satisfying properties (i) and (ii), then there is a permutation π with $\text{inv}(\pi) = X$.
11. Prove the following properties of a Coxeter group W (if you didn't already yesterday)
 - (a) Strong Exchange Property: If $t \in \text{inv}(w)$ and $s_{i_1} \cdots s_{i_k}$ is a reduced word for w , then there exists a j so that $tw = s_{i_1} \cdots \widehat{s}_{i_j} \cdots s_{i_k}$. (The hat notation means to omit s_{i_j} from the word.)
 - (b) Exchange Property: If s is a descent of w and $s_{i_1} \cdots s_{i_k}$ is a reduced word for w , then there exists a j so that $ws = s_{i_1} \cdots \widehat{s}_{i_j} \cdots s_{i_k}$.
 - (c) Deletion Property: If $s_{i_1} \cdots s_{i_k}$ is a non-reduced word for w , then there exist j and j' so that $w = s_{i_1} \cdots \widehat{s}_{i_j} \cdots \widehat{s}_{i_{j'}} \cdots s_{i_k}$.
12. Define the support of the word $s_{i_1} \cdots s_{i_k}$ to be the set $\{s_{i_1}, \dots, s_{i_k}\}$. Show that any two reduced words for an element w in a Coxeter group W have the same support.
13. Prove that the weak order on the symmetric group S_n is a **lattice**: this means that there is a least upper bound and greatest lower bound for any $x, y \in S_n$. You can just show there are least upper bounds. A least upper bound, or **join**, for x and y is an element z_0 in $Z = \{z \in S_n \mid x \leq_R z \text{ and } y \leq_R z\}$ such that for any $z \in Z$, we have $z_0 \leq_R z$. (Bonus problem: prove the lattice property for weak order on any finite Coxeter group.)

14. There are two simple ways to transform a reduced word in S_n into a different reduced word. One is to use the fact that $s_{i_j} s_{i_{j+1}} = s_{i_{j+1}} s_{i_j}$ whenever s_{i_j} and $s_{i_{j+1}}$ commute, allowing us to swap those two simple reflections in the reduced word without changing its product. This is called a commutation move. The other is to use the fact that $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$, which allows us to replace a string of three consecutive letters matching the pattern $s_i s_{i+1} s_i$ or $s_{i+1} s_i s_{i+1}$ with the other pattern. This is called a braid move. Prove that any two reduced words for $w \in S_n$ can be transformed into each other by performing a sequence of commutation and braid moves. (This is called Matsumoto's Theorem.)